Enhancing Thai Text Clustering for Performing Arts Through Domain-Specific NLP and Clustering Algorithms

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***Abstract*— This study aims to enhance the accuracy of Thai text clustering. It focuses specifically on text related to Thai performing arts. The objective is to improve information retrieval and support the preservation of cultural heritage. The research focuses on improving text preprocessing, embedding techniques, and clustering algorithms. The preprocessing pipeline includes word segmentation, part-of-speech tagging, stopword removal, and noun extraction, with domain-specific dictionaries incorporated for improved accuracy. Various embedding methods, including FastText, bidirectional encoder representations from transformers (BERT), WangchanBERTa, and BGE-M3, were evaluated. Dimensionality reduction techniques, such as principal component analysis (PCA) and uniform manifold approximation and projection (UMAP), were employed to optimize clustering performance. Among the clustering methods tested—k-means, DBSCAN, hierarchical clustering, spectral clustering, and gaussian mixture models (GMM)—k-means with UMAP achieved the highest Silhouette score (0.7115), indicating well-defined clusters. Additional evaluation using Davies-bouldin index (0.3948) and Calinski-harabasz index (2297.5827) confirmed the effectiveness of this approach. The resulting clusters were semantically coherent and thematically distinguishable, comprising groups related to folk performance, rituals and cultural practices; identity, symbolism and social values; mythology, royal traditions and sacred narratives; and deities, warriors and legendary figures.**

Expert validation further ensured the semantic coherence of clustered terms. These findings contribute to the advancement of natural language processing (NLP) methodologies for Thai text and can be applied in AI-driven recommendation systems and digital cultural heritage preservation.

Keywords*— *Thai performing arts, text clustering, natural language processing, word segmentation, embeddings

I. Introduction

In the era of big data, personalized recommendation systems powered by artificial intelligence (AI) have become essential tools for filtering large volumes of information and delivering content tailored to user preferences [1]. These systems help alleviate the problem of information overload, increasing user satisfaction and engagement [2], while also supporting customer retention and boosting revenue across various industries[3]. However, excessive recommendations can backfire, resulting in cognitive overload, where decision-making becomes more difficult due to dispersed or excessive information [4] .

To address this burden for both users and systems, one widely adopted solution is text clustering, which groups documents or words with similar meanings into clusters. This technique helps reduce cognitive and computational load [5], and also improves the accuracy and efficiency of recommendation systems[6] [7]. Traditional clustering approaches which often use basic feature extraction methods such as term frequency, document frequency and bag of words, along with algorithms such as k-means, hierarchical clustering, fuzzy c-means, and DBSCAN[8][9] which tend to underperform when dealing with short texts or data with deep contextual dependencies, as they are unable to capture nuanced meanings or varied contexts [10][11] .

Recent advancements in natural language processing (NLP) have introduced word embedding techniques such as Word2Vec and GloVe [12] , which encode semantic relationships in dense vector formats. Although these models outperform traditional vector representations, they are static and cannot adapt to changes in meaning when the same word appears in different contexts[13]. The emergence of large language models (LLMs), such as BERT (bidirectional encoder representations from transformers) has significantly enhanced the ability to generate context-aware embeddings, making them highly suitable for complex clustering tasks [14]. Deep learning architectures like autoencoders have also been applied to reduce the dimensionality of textual data, further improving clustering quality[15]. However, utilizing LLMs typically requires substantial computational resources and often involves tuning many parameters [16].

Despite the growing success of LLM-based embeddings, the preprocessing stage remains a critical step before feeding text into the models which especially for the Thai language, which lacks clear word boundaries. Word segmentation is a major challenge, as standard tools may struggle with domain-specific terms or out-of-vocabulary words, degrading the quality of embeddings and clustering results[17]. Therefore, this research proposes a data preparation approach specifically tailored for a corpus of Thai performing arts. The process includes custom word segmentation, part-of-speech (POS) tagging, stopword filtering, and noun phrase extraction to produce clean and context-relevant text inputs for more effective embedding and clustering.

This issue is particularly evident in the domain of Thai performing arts, which features unique vocabulary rarely found in standard Thai corpora[18][19]. For example, the term "Khorlor", a percussion instrument used in traditional Northeastern ensembles, may be wrongly segmented into "khor" (meaning “to request”) and "lor" (a meaningless term). This missegmentation alters the intended meaning and can result in incorrect clustering. Standard segmentation tools, therefore, fall short in such culturally specific contexts, highlighting the need for a customized approach.

Given this context, the objective of this study is to develop a Thai-specific data preparation pipeline and suitable

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embedding techniques to enable high-performance text clustering in personalized recommendation systems. The work focuses on the domain of Thai performing arts, which contains culturally specific terms and contexts not typically found in standard datasets. By integrating natural language processing and advanced clustering methods into this specialized domain, the proposed approach aims to improve recommendation accuracy, support cultural preservation and promote the sustainable dissemination of Thai performing arts.

The main contributions of this study are as follows:

1. The development of a domain-specific Thai text preprocessing pipeline, which incorporates customized word segmentation, POS tagging, stopword filtering, and noun extraction tailored for Thai performing arts.
2. A comparative evaluation of embedding techniques (FastText, BERT, WangchanBERTa, and BGE-M3) and clustering algorithms (k-means, DBSCAN, hierarchical, spectral, and GMM), as implemented in the proposed model.
3. The integration of dimensionality reduction techniques (PCA and UMAP) to improve clustering quality, leading to an efficient Thai text clustering framework applicable to personalized recommendation systems and cultural content analysis, which aligns with the proposed model and is validated through the results and discussion.

The remainder of this paper is structured as follows. Section II reviews related work on Thai text preprocessing, word embedding techniques, and clustering algorithms. Section III describes the proposed model, including domain-specific preprocessing, embedding strategies, and clustering methods. Section IV presents the experimental setup and results, highlighting the evaluation of preprocessing, embedding, and clustering performance. Finally, Section V concludes the study and outlines directions for future research.

II. RELATED WORK

This section reviews prior studies related to Thai text clustering and personalized recommendation systems within the context of natural language processing (NLP) and cultural applications. The discussion is structured into two focus areas that align with the key components of the proposed framework. The first area, text preprocessing, addresses fundamental steps such as word segmentation, part-of-speech tagging, stopword filtering, and noun extraction, which are essential for preparing Thai text for downstream tasks. The second area, text clustering, covers methods for representing text as numerical vectors, applying dimensionality reduction techniques and employing clustering algorithms to group semantically similar terms.

A. Text preprocessing

Text preprocessing ensures data quality, organization, and suitability for analysis, significantly enhancing clustering accuracy [23]. Thai text preprocessing typically includes word segmentation, part-of-speech (POS) tagging, stopword removal, and noun extraction.

1) Word Segmentation: Thai lacks spaces between words, necessitating segmentation methods. Damrongkamoltip et al. (2024) evaluated Thai segmentation techniques using LST20, finding Attacut superior in recall (0.90) and F1-score (>0.90). Deepcut and Newmm also performed well, with precision values exceeding 0.94. Regarding speed, Newmm was the fastest, making it suitable for real-time applications[24].However, traditional methods like Newmm struggle with out-of-vocabulary (OOV) words [25]. Learning-based models like Deepcut and Attacut improve accuracy but require extensive training data. Hybrid approaches combining dictionary-based methods with contextual embeddings have shown promise [26]. Our approach integrates domain-specific dictionaries and multiple embeddings, including FastText, WangchanBERTa, and BGE-M3, to enhance segmentation accuracy for Thai performing arts.

2) Part-of-Speech (POS) Tagging: POS tagging classifies words into grammatical categories, crucial for text structure understanding and clustering accuracy[26]. Using orchid corpus and lst20 has significantly improved POS tagging in Thai NLP[17].

3) StopWord Removal: Removing frequently occurring but semantically insignificant words reduces data size and enhances computational efficiency [17] . Studies indicate that stopword removal refines keyword concentration and extraction accuracy [12][27] .

4) Noun Extraction: Nouns represent key concepts such as entities and locations, making noun extraction essential for clustering [26]. Extracting nouns improves clustering accuracy by emphasizing core document content [28]. Research shows that noun extraction enhances feature selection and reduces data dimensionality while improving clustering efficiency [20][23].

B. Text clustering

Text clustering uncovers patterns in large text corpora, improving data organization and analysis [11]. The main steps include text representation, dimensionality reduction, and clustering algorithms.

1) Text Representation: After preprocessing, text is transformed into numerical representations for analysis [3]. Common Thai embedding models include FastText [23], BERT [8], and WangchanBERTa [13], the latter being a Thai-specific contextual model trained on large-scale local corpora. BGE-M3[29] , a multilingual model supporting over 100 languages, demonstrates superior performance on Thai tasks and long-text handling compared to BERT-based models. We evaluate multiple embedding techniques to identify the most effective representation for subsequent clustering.

2) Dimensionality Reduction: Reducing feature dimensions enhances clustering efficiency while preserving critical information [30]. Principal component analysis (PCA) reduces dimensionality by maximizing variance, while uniform manifold approximation and projection (UMAP) better preserves both local and global structures, improving clustering outcomes [31].

3) Clustering Algorithms: K-means is widely used for efficiency and minimizing intra-cluster variance [32]. DBSCAN [6] effectively handles non-spherical clusters and

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noise. Hierarchical clustering constructs a hierarchy without requiring predefined clusters [9]. Spectral clustering leverages graph-based techniques for complex structures [11]. Gaussian mixture models (GMMs) probabilistically model clusters with the expectation-maximization (EM) algorithm[33] .

Various dimensionality reduction techniques, such as UMAP and PCA, have been explored in previous research for clustering applications. The selection of an optimal method depends on the dataset characteristics, which will be further investigated in future work. These enhancements improve efficiency and accuracy in clustering Thai performing arts data, facilitating better text analytics and organization.

III. PROPOSED MODEL

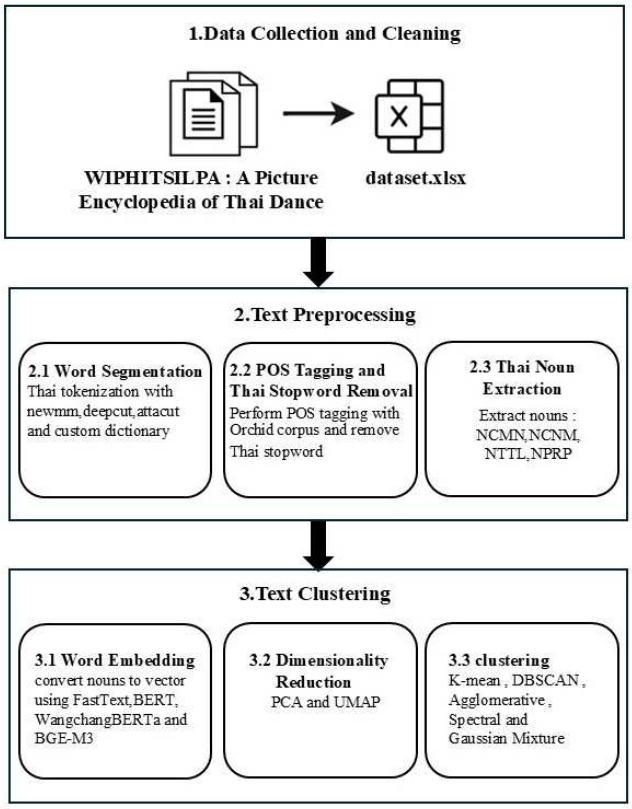

Fig. 1. Diagram of the Thai performing arts text clustering pipeline

From Fig. 1, The proposed model enhances Thai text clustering in the domain of performing arts through three main stages: (1) data collection and cleaning, (2) text preprocessing, and (3) text Clustering. It integrates techniques such as word segmentation, part-of-speech tagging, noun extraction, word embeddings, and dimensionality reduction before applying clustering algorithms. The model employs k-means, DBSCAN, agglomerative clustering, spectral clustering, and gaussian mixture to optimize clustering accuracy. This approach improves text organization and supports applications in recommendation systems and cultural heritage preservation.

A. Text preprocessing

This module prepares Thai text data for clustering by reducing noise and structuring data into a standardized format. It consists of four key steps: (1) data collection and preparation, (2) word segmentation, (3) POS tagging and stopword filtering, and (4) noun extraction.

1) Data collection and preparation: The dataset is sourced from Wiphitsilpa : A Picture encyclopaedia of Thai dances, published by the Bunditpatanasilp Institute of Fine Arts, Ministry of Culture, and the Thailand Foundation. The data, provided in spreadsheet format, undergoes cleaning to remove duplicates and missing values. Text normalization is applied to ensure consistent character formatting and spacing for subsequent processing. The Thai Words Corpus is used as the ground truth for evaluation in the segmentation step. A custom dictionary enhances accuracy by incorporating domain-specific terms from Thai performing arts, preventing segmentation errors. The construction of this dictionary involved manually curating over 200 culturally specific terms extracted from academic literature, encyclopedias, and glossaries related to Thai performing arts. This process required expert validation and iterative refinement, contributing significantly to segmentation performance while also introducing considerable manual workload.

2) Word segmentation: Thai text requires segmentation as it lacks explicit word boundaries, with descriptive data of the performance sets serving as input. This study evaluates three techniques: Newmm, which uses a maximum matching algorithm for dictionary-based segmentation; Deepcut, which employs a BiLSTM-based neural network for boundary prediction; and Attacut, which implements an attention-based segmentation mechanism. Newmm may struggle with out-of-vocabulary (OOV) words, while Deepcut and Attacut improve accuracy but require large training datasets. Segmentation performance is measured using precision, recall, and F1-score to evaluate accuracy and effectiveness.

3) POS tagging and Thai stopword filtering: After segmentation, POS tagging is performed using the orchid corpus to classify words into categories such as nouns, verbs, and adjectives. Additionally, Thai Stopword Filtering removes common words that do not contribute significant meaning.

4) Noun extraction: Noun extraction focuses on identifying key terms relevant to meaning representation. Words categorized under NCMN (Common Nouns), NCNM (Proper Nouns), NTTL (Titles), and NPRP (Pronouns) are selected to form a refined dataset for clustering, ensuring that only the most meaningful features are used in further analysis.

B. Text clustering

Text clustering structures textual data by grouping semantically similar words, enhancing information retrieval and supporting applications like recommendation systems and cultural heritage preservation. The process consists of four steps: word embedding, dimensionality reduction, clustering, and evaluation.

1) Word embedding: Extracted nouns are transformed into numerical representations using different embedding techniques: FastText, which is subword-based and effective for Thai text; BERT-Multilingual, which provides contextual embeddings across multiple languages; WangchanBERTa, a Thai-specific BERT model offering enhanced contextual

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understanding; and BGE-M3 Embedding, optimized for high-performance information retrieval. These embeddings are evaluated using clustering quality metrics to determine the most effective representation for Thai performing arts data. The embedding model with the best performance will be selected for use in the following dimensionality reduction and clustering steps.

2) Dimensionality reduction: Reducing dimensionality enhances clustering efficiency. Two techniques are compared: PCA (Principal Component Analysis), which captures global variance while minimizing data loss, and UMAP (Uniform Manifold Approximation and Projection), which preserves both local and global structures, crucial for high-dimensional embeddings. The selection of the optimal technique is based on clustering performance, measured by intra-cluster cohesion and inter-cluster separation. Dimensionality reduction will be applied to the embeddings generated by the model that achieves the best performance in the previous step.

3) Clustering: Multiple clustering algorithms are evaluated, including k-means, a centroid-based clustering method known for efficiency but sensitive to initial centroid selection; DBSCAN, a density-based clustering technique that effectively handles noise but requires fine-tuned parameter settings; Agglomerative clustering, which follows a hierarchical clustering approach that does not require predefined cluster numbers; Spectral clustering, which applies graph-based methods to detect complex structures; and Gaussian mixture model (GMM), a probabilistic clustering model suitable for non-spherical clusters. The elbow method and Silhouette score are used to determine the optimal number of clusters, which guide the parameter settings in the experiments.

C. Evaluation

Clustering quality is assessed using three metrics: Silhouette score, where higher values indicate better-defined clusters by measuring intra-cluster cohesion; Davies-bouldin index, where lower values suggest well-separated clusters by evaluating compactness and separation; and Calinski-harabasz index, where higher values reflect a well-structured clustering arrangement.

D. Limitations and Computational Considerations

Despite the effectiveness of the proposed approach, several limitations were observed. Firstly, the Thai language's lack of explicit word boundaries and the presence of out-of-vocabulary (OOV) terms—especially within niche cultural domains—posed challenges for standard segmentation and POS tagging techniques. Although mitigated by the custom dictionary, segmentation errors may still persist and influence downstream clustering quality.Moreover, The combined use of high-dimensional embeddings (e.g., WangchanBERTa, BGE-M3) and complex dimensionality reduction algorithms (e.g., UMAP) imposes considerable computational demands, particularly during training and evaluation phases. While such setups typically require GPU resources, the relatively compact sizes of these models—approximately 1GB for WangchanBERTa and 2.3GB for BGE-M3—make them scalable on standard GPU hardware without the need for high-end servers. This indicates that the approach remains feasible for practical deployment in most environments. Nevertheless, future work may still explore lightweight or quantized alternatives to improve efficiency and deployment viability, especially in large-scale or real-time applications.

IV. RESULT AND DISCUSSION

This section presents the findings and analysis of text preprocessing and clustering performance, providing insights into the effectiveness of various methods. The experiments were conducted on a notebook optimized for nlp tasks, particularly deep learning-based word embeddings and clustering. The experimental setup was designed to handle Thai text segmentation, embedding and clustering. It ensures both efficiency and reproducibility.

A. Experiment Setup

Hardware and Software: Experiments were conducted on a notebook selected for its efficiency in handling nlp tasks, particularly those involving deep learning-based word embeddings and clustering. The system was powered by an AMD Ryzen AI 9HX 370 processor with integrated Radeon 890M graphics, supported by 32GB of RAM, and operated on Windows 11 Home (24H2).

The programming environment was built using Python 3.11.8 and managed through PyCharm. Required packages were synchronized using the sync python requirements tool to ensure all dependencies were included accurately. The programming environment included several essential Python packages. We used pandas (v2.2.3) for structured data analysis and numpy (v1.26.4) for matrix operations and numerical computations. The pythainlp (v5.1.0) library was utilized for Thai text preprocessing, with Deepcut (v0.7.0.0) and Attacut (v1.0.6) serving as alternative segmentation models. Visualization tasks were handled using matplotlib (v3.10.1) and seaborn, while scikit-learn (v1.6.1) and umap-learn (v0.5.7) supported clustering and dimensionality reduction. Deep learning models were executed using torch (v2.6.0+cu118), with contextual embeddings provided by transformers (v4.49.0), which includes support for models like BERT and WangchanBERTa. Additional packages such as requests (v2.32.3) facilitated access to external APIs, while gensim (v4.3.3) and sentence-transformers (v3.4.1) supported vector-based and sentence-level embeddings, respectively. The scipy (v1.13.1) package was employed for scientific computation.

All dependencies were synchronized using pycharm's sync python requirements tool and exported to a requirements.txt file. This ensured a reproducible environment that could be easily replicated using the command pip install -r requirements.txt, supporting all stages of experimentation from preprocessing to clustering.

The hyperparameters used in this study were optimized to ensure efficient word embedding, dimensionality reduction, and clustering. As summarized previously in table I, these parameters were selected based on best practices from the literature and empirical tuning to maximize clustering performance.

For dimensionality reduction, PCA and UMAP were employed. The parameters for vector generation were set as vector_size=300, window=5, and min_count=1, which are commonly used values in word embedding tasks to balance vector dimensionality and semantic richness.

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For clustering, k-means was configured with n_clusters set to the optimal number determined using the elbow method. The random_state was fixed at 42 to ensure reproducibility, and n_init was set to 10 to avoid poor local minima from random initialization.

DBSCAN was configured with eps=0.5 and min_samples=5 to detect dense regions while remaining robust to noise which suitable for uncovering non-spherical clusters in the embedding space.

Agglomerative clustering and spectral clustering both set the number of clusters (n_clusters) based on k from prior evaluations. In the case of agglomerative clustering, it employs ward's linkage, which iteratively merges clusters in a way that minimizes the increase in the total within-cluster variance. Essentially, at each step, the algorithm selects the pair of clusters whose union results in the smallest possible increase in variance, leading to clusters that are as compact and internally similar as possible.

GMM used n_components equal to the number of target clusters to model the data as a mixture of Gaussians, accommodating potential overlaps in semantic space.

These settings, informed by table I, were essential for establishing a reliable and consistent experimental pipeline across all clustering models.

B. Data preparation

Before segmentation, the raw dataset underwent a cleaning process to remove duplicates and entries with missing values. This step also included character normalization, whitespace trimming, and the standardization of punctuation. As a result, the dataset was transformed into a cleaner and more consistent format, ensuring high-quality input for subsequent nlp processes. This preparation significantly reduced noise in the data, which contributed to more accurate segmentation and improved overall model performance.

TABLE I. HYPERPARAMETERS CONFIGURATIONS

Process	Parameter	Value
Dimensionality Reduction	PCA, UMAP	vector_size=300,window=5, min_count=1
Clustering	K-Means	n_clusters=optimal_k (determined using elbow method), random_state=42, n_init=10
	DBSCAN	eps=0.5, min_samples=5
	Agglomerative	n_clusters=num_clusters, linkage='ward'
	Spectral	n_clusters=num_clusters
	GMM	n_components=num_clusters

TABLE II. SEGMENTATION PERFORMANCE METRICS

Method	Avg Processing Time (s)	Precision	Recall	F1-score
Newmm	0.000233	0.956141	0.000746	0.001491
Deepcut	0.090135	0.907748	0.000742	0.001483
Attacut	0.004286	0.908772	0.000744	0.001486

- Newmm : กินเนสส์บุ๊ค
- Deepcut : กินเนสส์ | บุ๊ | ค
- Attacut: กิน | เนสส์บุ๊ค

Fig. 2. Example of word segmentation results using Newmm, Deepcut, and Attacut.

C. Text preprocessing

1) Word Segmentation Performance: The performance of Newmm, Deepcut, and Attacut segmentation techniques was evaluated using the thai words corpus (thai_words()) as the golden standard. Precision, recall, and F1-score were used as evaluation metrics. The evaluation results are shown in table II

From table II, Newmm achieved the highest precision and the lowest processing time but exhibited low recall due to its dictionary-based approach, which struggled with out-of-vocabulary (OOV) words. Deepcut and Attacut showed improved handling of OOV words but introduced segmentation errors in domain-specific terms. A custom dictionary was used to mitigate segmentation errors by incorporating domain-specific terms related to Thai performing arts, thereby improving segmentation accuracy. Examples of these segmentation errors are illustrated in Fig. 2, which highlights domain-specific segmentation challenges.

2) POS tagging and Stopword Filtering: POS tagging was performed using the ORCHID corpus, categorizing tokens into nouns, verbs, and adjectives. Thai stopwords were removed to reduce noise. As shown in table III, this process reduced nouns from 2659 to 1079, verbs from 668 to 267 and adjectives from 98 to 46.

3) The noun extraction process aimed to identify key terms essential for meaning representation, with a focus on common nouns, proper nouns, and titles. These noun types were selected based on their semantic relevance to enhance clustering quality.

Examples of these extracted noun types are shown in Fig. 3. These extracted nouns contribute to improved clustering performance by ensuring that semantically relevant features are prioritized during vectorization and grouping.

D. Text clustering

1) Word *embedding:* To enable effective text clustering, Thai noun phrases were transformed into numerical vectors using four embedding models: FastText, BERT (via SentenceTransformer), WangchanBERTa, and BGE-M3. The embedding vectors were then reduced to two dimensions using t-SNE for visualization and comparison.

- Common Nouns (NCMN): ประเพณี,วัฒนธรรม,การรำ
- Proper Nouns (NPRP): พระราม,พระลักษณ์,พระนารายณ์
- Titles (NTTL): พระเจ้าชัยวรมัน

Fig. 3. Examples of extracted common nouns, proper nouns, and titles

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Algorithm 1: Thai Noun Embedding and Visualization

Input: Noun phrases extracted from Thai text corpus
Output: 2D embedding visualizations for comparison

Load noun phrases from input Excel file
Preprocess to retain unique Thai words only

// Generate embeddings from four models
For each word in noun list:
    Generate FastText vector
    Generate BERT vector (SentenceTransformer)
    Generate WangchanBERTa vector
    Generate BGE-M3 vector via API

// Reduce dimensions for visualization
For each embedding set:
    Apply t-SNE to reduce vectors to 2D
    Plot the 2D vectors using scatter plot

Compare cluster compactness across models
```

Fig. 4. Key pseudo code for word embedding

From Fig. 4, the pseudocode for the embedding process is illustrated in a step-by-step manner. The process begins with loading Thai noun phrases extracted from the text corpus, followed by preprocessing to retain only unique Thai words. Then, word embeddings are generated for each word using four different models: FastText, BERT (via Sentence Transformer), WangchanBERTa, and BGE-M3 (via API). Afterwards, the high-dimensional vectors are reduced to two dimensions using t-SNE for each model. Finally, the 2D vectors are plotted using scatter plots to visually compare the distribution and compactness of clusters across models, as shown in Fig. 5.

From Fig. 5, the 2D visualization of Thai performing arts terms from each model is presented. FastText and BERT embeddings exhibit dense, unstructured distributions. BGE-M3 shows a slightly wider spread, though still centralized. In contrast, WangchanBERTa reveals well-separated clusters, indicating better semantic grouping.

TABLE III. POS TAGGING STATISTICS

POS Category	Before Stopword Removal	After Stopword Removal
Nouns (NCMN, NCNM, NTTL, NPRP)	2659	1079
Verbs (VACT, VSTA)	668	267
Adjectives (VATT)	98	46

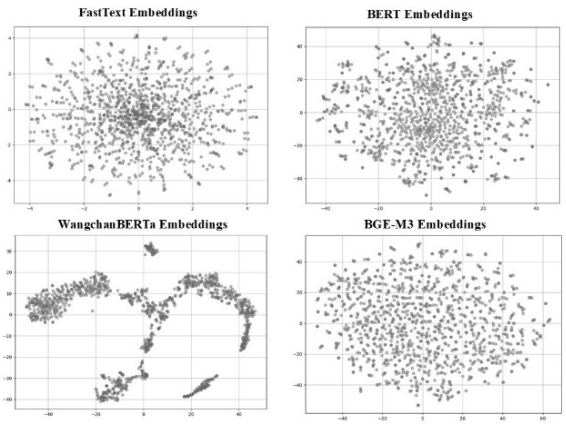

Fig. 5. Word embedding visualization for Thai performing arts terms

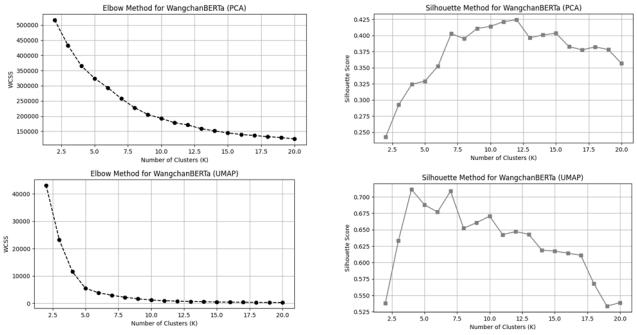

Fig. 6. Dimensionality reduction performance of PCA and UMAP

These results suggest that WangchanBERTa provides more distinguishable representations for Thai terms, making it a promising choice for downstream clustering and recommendation tasks in the cultural domain.

2) Dimensionality Reduction Performance

Dimensionality reduction is a crucial step in simplifying the complexity of word embedding vectors. In this study, we examined dimensionality reduction techniques applied to vectors generated by WangchanBERTa, including PCA and UMAP.

From Fig. 6, Dimensionality reduction performance, it is evident that UMAP outperforms PCA in reducing dimensionality for clustering purposes, as follows:

- Elbow method (PCA vs. UMAP): The elbow method graph for PCA shows a continuous decline in WCSS as the number of clusters (K) increases, making it difficult to identify a clear optimal K. In contrast, UMAP presents a distinct elbow point at approximately K = 4 to 5, indicating a well-structured cluster distribution.
- Silhouette score (PCA vs. UMAP): The graph illustrates that PCA achieves its highest Silhouette score of approximately 0.42 when K is between 10 and 12, whereas UMAP attains a significantly higher maximum Silhouette score of approximately 0.71 at k = 4 to 5. This suggests that clustering using UMAP results in well-separated clusters with high intra-cluster similarity, indicating superior group coherence compared to PCA.

3) Clustering and evaluation

We evaluated word embedding clustering using various techniques, including k-means, DBSCAN, hierarchical clustering, spectral clustering, and gaussian mixture models (GMM), applied to vectors reduced using UMAP.

These results suggest that WangchanBERTa provides more distinguishable representations for Thai terms, making

TABLE IV. CLUSTERING PERFORMANCE EVALUATION

Method	Optimal K	Silhouette score	Davies-bouldin index	Calinski-harabasz index
K-Means	**4**	**0.7115**	**0.3948**	**2297.5827**
DBSCAN	-	0.5795	0.4546	2533.0276
Hierarchical	**4**	**0.7115**	**0.3948**	**2297.5827**
Spectral	4	0.1983	1.3118	254.9959
GMM	4	0.7094	0.4005	2296.9980

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it a promising choice for downstream clustering and recommendation tasks in the cultural domain. Optimal number of clusters selection was performed using the elbow method and Silhouette score. The results indicate that k-means (UMAP vectors) and hierarchical clustering (UMAP vectors) performed best at K=4, achieving the highest Silhouette score of 0.7115 and the lowest Davies-bouldin index of 0.3948 as shown in table IV.

From table IV, the results demonstrate that K-Means and Hierarchical Clustering using UMAP yielded the best performance, effectively distinguishing word clusters. In contrast, Spectral Clustering and DBSCAN struggled due to the complexity of the data and failed to generate well-separated clusters.

From the data in table IV, k-means and hierarchical clustering achieved the same performance scores in terms of Silhouette score and Davies-bouldin index. Therefore, the decision on the most appropriate clustering method may require additional visual inspection, such as analyzing Figure 3: comparison of k-means and hierarchical clustering on UMAP 2D representation.

Based on the analysis of Fig. 7, the UMAP visualization shows that K-Means produces well-separated and evenly distributed clusters, while Hierarchical clustering tends to merge nearby groups into larger clusters, resulting in less distinct boundaries. Although both methods yield similar numerical performance, K-Means demonstrates superior clarity in cluster structure, making it more suitable for tasks requiring clearly defined and non-overlapping groups.

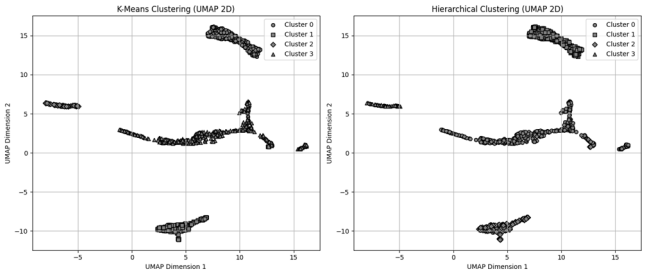

Fig. 7. Comparison of k-means and hierarchical clustering on UMAP 2D representation

Additionally, the word clustering results from k-means at k=4 exhibit clear thematic groupings, reflecting different aspects of Thai cultural, artistic, and mythological concepts. Cluster 0 consists mainly of words related to traditional performances and historical references, cluster 1 emphasizes performing arts and musical instruments, cluster 2 captures elements of symbolic and aesthetic significance, while cluster 3 focuses on themes of warriors, deities, and battles. These findings suggest that k-means clustering effectively groups words into semantically meaningful categories, demonstrating its applicability for semantic analysis in Thai linguistic studies.

V. CONCLUSION

This study demonstrates the feasibility and effectiveness of using AI-based text clustering techniques to analyze and categorize Thai performing arts data. Among the evaluated models, the combination of WangchanBERTa embeddings, UMAP dimensionality reduction, and k-means clustering achieved the highest clustering performance, with a silhouette score of 0.7115.

The experimental results yielded four semantically coherent clusters, each representing a distinct thematic category within Thai cultural terminology. Cluster 0 comprised terms related to folk performance, rituals and cultural practices. Cluster 1 focused on identity, symbolism and social values. Cluster 2 represented mythological themes, royal traditions and sacred narratives. Cluster 3 included deities, warriors and legendary figures. These well-separated groupings highlight the model's ability to capture nuanced domain-specific semantics within the context of Thai performing arts.

The proposed system holds strong potential for use in AI-driven recommendation systems for Thai performing arts, enhancing the precision, relevance, and cultural richness of suggestions. Future enhancements, including the integration of adaptive NLP pipelines for real-time applications and the use of lightweight or quantized models, could improve scalability and responsiveness. Additionally, expanding the dataset to encompass broader aspects of Thai performing arts and implementing automatic content generation could further promote the digital preservation and global appreciation of Thai cultural heritage.

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